

EBAI BATE LUCKY BETTY

SOFTWARE ENGINEER · CREATIVE TECHNOLOGIST

Application Developer

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Yaoundé, Cameroon

Open to International & Remote

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lucky-platform-five.vercel.app

github.com/Ebai-Bate-Lucky-Betty/lucky-platform

PROFILE

Software Engineer and Computer Science researcher focused on building reliable applications, intelligent systems, and interactive digital experiences. Experienced in backend and application development, distributed architectures, microservices, machine learning, and explainable AI.

I combine engineering with a strong creative practice spanning interactive experiences, immersive technology, 3D environments, storytelling, and world-building, exploring how technology can transform ideas and narratives into interactive experiences.

Currently pursuing a research-oriented Master's degree in Computer Science, with interests in intelligent systems, software testing, explainability, and distributed applications.

EDUCATION

Master's Degree in Computer Science, Research Track (SIGL) | 2024 – Present

University of Yaoundé I, Yaoundé, Cameroon

- **Research focus:** Explainable AI, software testing, microservices, execution-trace analysis, and intelligent test prioritization.
- **Current research topic:** Explainable Risk Prediction for Test Path Prioritization in Microservices: A SHAP-Based Approach Using Execution Traces
- **Relevant areas:** Machine Learning · Explainable AI · Software Engineering · Distributed Systems · Microservices · Symbolic Execution

Bachelor's Degree in Computer Science | 2021 – 2024

University of Yaoundé I, Yaoundé, Cameroon

PROFESSIONAL EXPERIENCE

Software Development Intern – KB Group, Yaoundé | 4 Months

- Contributed to the development of **Skillcert**, a platform designed to support knowledge assessment and testing.
- Participated in application development and implementation of software features within a team environment.
- Worked with software development practices involving application logic, interfaces, testing, and deployment considerations.
- Gained practical experience translating functional requirements into working software.

SELECTED PROJECTS

Fraud Detection for Document Authenticity

Machine Learning · Computer Vision · Explainable AI · Python

A document-authenticity detection system designed to distinguish authentic documents from manipulated or fraudulent documents.

- Built a supervised image-classification pipeline using the **SIDTD** dataset.
- Developed a reproducible dataset indexing and group-aware splitting workflow.
- Implemented a baseline image-classification model using **EfficientNet-B0**.
- Designed the system to classify documents into **authentic, suspicious, and fraudulent** outcomes.
- Integrated **SHAP** to explain model predictions and make classification decisions more interpretable.
- Structured the project toward future organisational use cases, including custom model training from organisation-specific datasets and document metadata.

Technologies : Python · PyTorch · Torchvision · EfficientNet · SHAP · OpenCV · Machine Learning

DreamHouse237 – Microservices Platform

Microservices · Backend Engineering · Distributed Systems

A distributed application developed as a team project using a microservices architecture.

- Contributed to the **architecture and design of the overall microservices system**.
- Designed and developed the **User Service**.
- Designed and developed the **Identity/Identification Service**, including document-based identity verification workflows.
- Worked with OCR technologies for extracting information from identity documents.
- Contributed to service communication and integration using **RabbitMQ**.
- Worked with **Spring Cloud Gateway** and **Eureka** for API routing and service discovery.
- Worked within an architecture involving authentication, user management, identity verification, publication, commentary, and payment services.
- Contributed to the design of a distributed architecture involving cloud-hosted infrastructure and databases.

Technologies : Java · Spring Boot · Spring Cloud · RabbitMQ · PostgreSQL · OpenCV · Tesseract OCR · AWS

Explainable Test Path Risk Prediction

Research Project · Explainable AI · Software Testing

Research project investigating machine-learning-based prioritisation of test paths in microservice systems.

- Designed a risk-prediction approach using execution traces collected from microservice applications.
- Investigated features such as service-call depth, branch depth, historical errors, and messaging behaviour.
- Exploring **Random Forest** and **XGBoost** models for risk prediction.
- Using **SHAP** to explain the factors influencing predicted test-path risk.
- Investigating **symbolic execution with Angr** to enrich path-level analysis.
- Evaluating prioritisation using **Precision@K, PR-AUC, and F1-score**.
- Comparing ML-based prioritisation with traditional strategies: random, DFS, BFS, and depth-based.

Technologies : Python · XGBoost · Random Forest · SHAP · Angr · Microservices · Machine Learning

TECHNICAL SKILLS

Programming : Python · Java · JavaScript · TypeScript · SQL

Backend & Application Development : Django · Django REST Framework · Spring Boot · Next.js

Distributed Systems : Microservices · REST APIs · Service Discovery · API Gateways · RabbitMQ · Distributed Architecture

Data & Machine Learning : PyTorch · Computer Vision · Scikit-learn · XGBoost · SHAP · OpenCV · Tesseract OCR

Databases : PostgreSQL · MariaDB · MySQL

DevOps & Tools : Git · GitHub · Docker · Linux · Vercel

Creative Technology : Blender · 3D Environments · Interactive Experiences · Immersive Technology · Digital Storytelling · Narrative Design · World-Building

AI & Development Practices : Prompt Engineering · Explainable AI · Model Evaluation · Software Testing · Research Prototyping

RESEARCH INTERESTS

- Explainable Artificial Intelligence
- Intelligent Software Testing
- Machine Learning for Software Engineering
- Microservices and Distributed Systems
- Symbolic Execution
- Computer Vision
- Intelligent Application Systems
- Immersive and Interactive Technologies
- Human-Centred Digital Experiences

CREATIVE TECHNOLOGY & STORYTELLING

Building a personal body of work at the intersection of **software engineering, creative technology, and storytelling**.

- Story writing and narrative development
- Creative scenario development
- Character and fictional-world development
- Interactive storytelling
- World-building and imaginative environments
- 3D environments and digital worlds
- Interactive and immersive experiences
- Game and experience concepts
- Visual and narrative concept development
- Exploring how stories can become interactive digital experiences

Creative direction : *Software Engineering × Intelligent Systems × Interactive Experiences × Storytelling × World-Building*

PROFESSIONAL STRENGTHS

- Software architecture and system thinking
- Backend and application development
- Problem decomposition and technical research
- Building end-to-end software projects
- Working across software engineering and machine learning
- Translating ideas into functional prototypes
- Technical documentation and communication
- Narrative thinking and creative concept development
- Independent learning and research

LANGUAGES

English : Professional proficiency

French : Professional proficiency